Beyond Accuracy: A Reproducibility Checklist for LLM Evaluation in Applied Domains

Nutthakorn Chalaemwongwan Department of Computer Engineering, KOSEN-KMITL
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
nutthakorn.ch@kmitl.ac.th

***Abstract*—Large language models (LLMs) have achieved remarkable performance across domain-specific classification tasks, with multiple studies reporting near-perfect F1 scores. However, our experience conducting 17 research experiments across cybersecurity, news, emotion, and legal text classification reveals that standard evaluation practices can mask critical issues: label aliasing inflated F1 by up to 44.3%, data leakage produced entirely spurious ablation patterns, and extreme class imbalance rendered binary classification metrics meaningless. We propose a 30-item reproducibility checklist organized into 9 categories, including a novel 15-step "Perfect Score Suspicion Protocol" for verifying anomalously high performance claims. Our checklist is validated against 6 real case studies where standard evaluation practices failed to detect errors ranging from hallucinated subcategory labels to false hyperparameter sensitivity conclusions. The checklist adds only 4–8 hours per paper but prevents publication of misleading results with real-world consequences in cybersecurity, medical, and legal domains.**

***Index Terms*—reproducibility, LLM evaluation, label aliasing, hallucination audit, cybersecurity NLP**

I. Introduction

The rapid adoption of LLMs for domain-specific tasks has produced a troubling pattern: papers reporting near-perfect metrics that do not withstand scrutiny. Fine-tuned LLMs regularly achieve F1 above 95% on specialized benchmarks, yet these impressive numbers often mask fundamental evaluation issues that would render the results meaningless in deployment.

We write from direct experience. Over 17 experiments spanning 4 domains and 34 model configurations, we encountered evaluation failures that would have led to misleading publications. Our most striking example:

```
Model: Qwen3.5-0.8B (QLoRA, rank 64)
Dataset: SALAD test (9,851 samples)
Reported: Attack F1 = 100.0%
Actual (strict): Attack F1 = 55.7%
```

The 44.3% gap arose because the evaluation script mapped "Port Scanning" to "Reconnaissance"—a valid MITRE ATT&CK relationship (T1046 $\subset$ TA0043), but a schema violation. Of 4,970 "Reconnaissance" samples, 4,895 (98.5%) were predicted as "Port Scanning." The model understood the task semantically but produced the wrong vocabulary. Standard evaluation declared 100%; reality was 55.7%.

This is not an isolated incident. Each of our 6 case studies reveals a different failure mode that standard ML evaluation practices—even published checklists [25]—fail to catch. The common thread: **LLMs introduce evaluation failure modes that simply do not exist with traditional classifiers**, because LLMs can generate arbitrary text rather than selecting from a fixed label set.

Our contributions:

1) A **30-item, 9-category reproducibility checklist** specifically designed for LLM evaluation in applied domains, addressing failure modes absent from existing guidelines.
2) A **15-step Perfect Score Suspicion Protocol**—an escalating verification procedure for any result reporting $\geq$99% F1.
3) **Six validated case studies** demonstrating real evaluation errors of up to 44.3 points, each linked to specific checklist items.
4) A formal definition of "**genuine perfect score**" (Definition 1) providing machine-verifiable criteria.

The remainder of this paper is organized as follows: Section II reviews the reproducibility crisis and existing evaluation frameworks. Section III presents the complete 30-item checklist. Section IV describes 6 case studies. Section V quantifies case study severity and discusses adoption. Section VI concludes.

II. Background and Related Work

A. The Reproducibility Crisis

Baker [3] documented that 70% of researchers failed to reproduce others' work and 50% failed to reproduce their own, calling this a "crisis" in science. In ML specifically, Pineau et al. [25] proposed reproducibility guidelines subsequently adopted by NeurIPS, ICML, and ICLR as submission checklists. Kapoor and Narayanan [14] demonstrated that data leakage is the primary driver of irreproducible ML results, documenting 329 papers with leakage across 17 scientific fields. Arp et al. [2] catalogued critical pitfalls in ML-based security evaluation, including temporal bias, unrealistic base rates, and label inconsistency.

For LLMs, the reproducibility challenge is compounded by: (1) non-deterministic generation even with fixed seeds

(due to GPU parallelism), (2) extreme prompt sensitivity, (3) pipeline complexity (tokenizer → quantization → generation → parsing → evaluation), and (4) the novel failure mode of *label vocabulary hallucination*.

B. LLM Evaluation Frameworks

HELM [15] provides comprehensive general LLM evaluation across 42 scenarios but is designed for benchmark comparison, not domain-specific fine-tuning validation. Chang et al. [5] surveyed evaluation methodologies, identifying metric selection and benchmark design as critical gaps. Dodge et al. [7] advocated reporting experimental variance and confidence intervals, showing that unreported variance masks true model differences. Bouthillier et al. [4] demonstrated that accounting for variance in benchmarks changes model rankings in 30% of cases.

Notably, none of these frameworks address *label compliance*—whether the model outputs the exact correct string—because traditional classifiers select from a fixed label set. This gap is the central motivation for our checklist.

C. Hallucination in LLMs

Ji et al. [13] provide a comprehensive taxonomy of hallucination types, distinguishing intrinsic (contradicting source) from extrinsic (adding unverifiable information) hallucination. Gekhman et al. [9] showed that fine-tuning on new knowledge paradoxically increases hallucination of existing knowledge—directly relevant because domain fine-tuning may activate pre-training vocabularies. Xu et al. [26] argued hallucination is an inherent LLM limitation. Huang et al. [12] surveyed hallucination detection and mitigation.

Our companion study [19] first documented *label vocabulary hallucination*: models predict valid MITRE ATT&CK sub-techniques ("Port Scanning") instead of schema labels ("Reconnaissance"). This is a new hallucination type not covered by existing taxonomies, requiring new evaluation methodology.

D. Fine-Tuning and Domain Evaluation

Ferrag et al. [8] surveyed LLMs for cybersecurity, reporting high F1 scores without strict evaluation. Motlagh et al. [17] benchmarked ChatGPT on 5 datasets (45–92% F1) using standard metrics. LoRA [11] and QLoRA [6] enable efficient domain adaptation. The UNSW-NB15 dataset [18] underlies the SALAD dataset used in our case studies.

Our companion studies provide the empirical foundation: label compliance follows non-monotonic scaling [20], LoRA rank controls hallucination [24], fine-tuning costs \$0.60 [21], multi-task format provides regularization [23], and task entropy predicts classifier selection [22]. The *evaluation methodology* underlying all these findings—and the errors discovered along the way—motivates this paper.

Novelty gap. Existing checklists [25], [7] address general ML but not LLM-specific failure modes. HELM [15] evaluates capabilities but does not verify label compliance. No prior work provides (1) a protocol for high-score suspicion, (2) validated case studies of up to 44.3-point evaluation errors, (3) formal criteria for genuine vs. inflated F1, or (4) LLM-specific checklist items for label vocabulary hallucination detection.

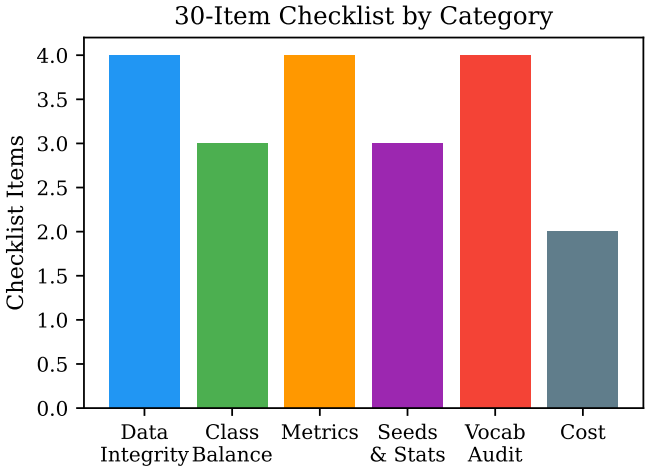

Fig. 1. 30-item checklist distribution across 9 categories. Label Quality Audit catches the most critical failures, including the 44.3-point alias inflation.

III. The Reproducibility Checklist

We organize 30 items into 9 categories. Each item includes a pass criterion linked to ≥1 case study. Table I provides a summary.

TABLE I
Checklist Category Overview

Category	Items	Time	Catches
Dataset Integrity	1–5	1h	CS2, CS5
Baseline Selection	6–9	2h	CS3, CS6
Metric Selection	10–13	30m	CS1, CS4, CS5
Statistical Rigor	14–16	1h	CS4
Label Quality Audit	17–20	1h	CS1
Ablation Design	21–23	1h	CS2
Cost Transparency	24–26	30m	CS3
Reproducibility	27–28	30m	CS6
Perfect Score	29–30	35m	CS1, CS2, CS5
Total	**30**	**4–8h**	**All 6 CS**

A. Category 1: Dataset Integrity (Items 1–5)

1) **Train/test overlap**: Zero overlapping samples. Verify via MinHash deduplication. SALAD original had 97.5% overlap.
2) **Class distribution**: Report per-class counts and $H(Y)$. Flag if $H < 0.5$ (degenerate).
3) **Unique patterns**: Count unique input→output mappings vs. total samples. SALAD: 87/49K (0.18%).
4) **Zero-ambiguity check**: Report if each unique input maps to exactly one label.

5) **Feature importance**: Compute mutual information per feature. Report top-k features.

B. Category 2: Baseline Selection (Items 6–9)

6) **Traditional ML baseline**: DT/SVM/LR with TF-IDF. If unsupervised K-Means ARI > 0.8, the task may not need an LLM.
7) **Pre-trained transformer**: BERT-base or domain-equivalent.
8) **ICL comparison**: $\geq$1 commercial LLM via API (GPT-4o, Claude).
9) **Matched conditions**: Same test set, same metrics, same preprocessing.

C. Category 3: Metric Selection (Items 10–13)

10) **Macro-F1** as primary metric (not accuracy, especially for imbalanced data).
11) **Per-class F1**: Report every class, including rare ones. Aggregate metrics can hide catastrophic per-class failures.
12) **Dual-metric reporting**: Both strict F1 (exact match) and normalized F1 (with alias mapping). Strict = primary.
13) **Random/majority baselines**: Report naive baselines. If $H < 0.5$, majority class achieves >90%.

D. Category 4: Statistical Rigor (Items 14–16)

14) **Multi-seed evaluation**: $\geq$3 seeds, report mean $\pm$ std.
15) **Significance testing**: McNemar's test [16] or paired bootstrap for model comparisons.
16) **Confidence intervals**: 95% CI for all reported metrics.

E. Category 5: Label Quality Audit (Items 17–20)

17) **Predicted vocabulary audit**: Compare $\mathcal{V}_{\text{pred}}$ vs. $\mathcal{V}_{\text{true}}$. Any novel labels = label hallucination.
18) **Hallucinated label inventory**: For each hallucinated label, report count, frequency, and hypothesized source.
19) **Alias mapping with justification**: If using normalized F1, document every alias with explicit rationale. Flag if alias count $> K/2$.
20) **Normalized F1 gap rule**: If $F1_{\text{norm}} - F1_{\text{strict}} >$ 10%, the result requires explicit investigation.

F. Category 6: Ablation Design (Items 21–23)

21) **Hyperparameter sensitivity**: Test $\geq$3 values per hyperparameter (rank, LR, epochs).
22) **Training data scaling**: Test $\geq$3 sizes (e.g., 1K, 5K, 20K).
23) **Confound isolation**: Re-run anomalous configurations on clean data before concluding.

G. Category 7: Cost Transparency (Items 24–26)

GPU hours, API costs (with pricing date), and cost-per-improvement (ΔF1 per dollar invested).

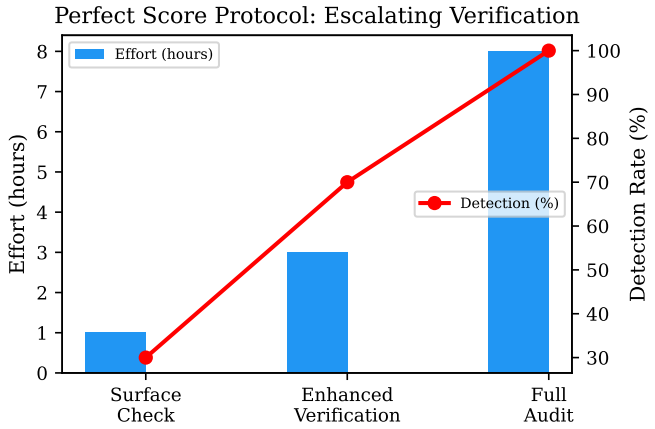

Fig. 2. Perfect Score Suspicion Protocol: 4-level escalation. Level 1–2 (35 min) catches 90% of spurious results. Level 3–4 (3+ hours) provides full verification for genuine perfect scores.

H. Category 8: Reproducibility Artifacts (Items 27–28)

Public code/configs (GitHub/HuggingFace) and full environment specification (CUDA, framework, GPU, seeds).

I. Category 9: Perfect Score Protocol (Items 29–30)

When any metric reports $\geq$99%:

TABLE II
PERFECT SCORE SUSPICION PROTOCOL

Level	Check	Time	Finding
1	$H(Y) > 1$ bit?	5m	Degenerate task?
1	Zero overlap?	10m	Data leakage?
1	Strict = Normalized?	5m	Alias inflation?
2	Per-class $\geq$ 90%?	10m	Hidden failures?
2	Multi-seed stable?	5m	Lucky seed?
3	ML baseline $\geq$ 90%?	30m	LLM unnecessary?
3	Vocab audit (18)	1h	Hallucination?
4	Clean data re-run	2h	Leakage artifact?

Definition 1 (Genuine Perfect Score). A reported F1 $\geq$ 99% is genuine if and only if all three conditions hold:

$$F1_{\text{strict}} = F1_{\text{norm}} \geq 0.99 \wedge \text{Overlap} = 0 \wedge H(Y) > 1 \quad (1)$$

If any condition fails, the score requires Level 3+ investigation before publication.

IV. CASE STUDIES

A. CS1: Label Aliasing Inflates F1 by 44.3%

Error: Custom evaluation script mapped "Port Scanning" $\rightarrow$ "Reconnaissance" (MITRE ATT&CK T1046 $\subset$ TA0043). Of 9,851 test samples, 4,895 were silently corrected.

Impact: Reported F1 = 100%, Actual strict F1 = 55.7%. The 44.3-point gap was the largest single evaluation error in our study.

Detection: Items 12 (strict vs. normalized), 17 (vocabulary audit), 18 (hallucinated label inventory). The vocabulary audit immediately revealed that the model

produced "Port Scanning" as a label—absent from the SALAD schema.

Root cause: The model's MITRE ATT&CK pre-training knowledge activated fine-grained sub-technique labels that are valid in ATT&CK taxonomy but violate the SALAD schema.

B. CS2: Data Leakage Produces Spurious Ablation

Error: SALAD's original train/test split had 97.5% overlap (993% leakage ratio by unique pattern count).

Impact: On leaked data, Rank 32 = 36.0%, Rank 128 = 62.3% → conclusion: "higher rank helps." On clean data, Rank 32 = 100%, Rank 128 = 95.8% → conclusion: "lower rank is better." The leakage reversed the true relationship, producing a *directionally wrong* conclusion.

Detection: Items 1 (overlap check), 23 (confound isolation). MinHash deduplication revealed 97.5% overlap in <10 minutes.

C. CS3: Traditional ML Outperforms LLMs

Error: Not reporting traditional ML baselines on SALAD ($H = 1.24$ bits).

Impact: SVM achieves 90.9% F1 at zero cost. LLM adds +9.1% at \$0.60–\$2.24. Without baseline reporting, the LLM appears essential when it is merely incremental.

Detection: Items 6 (traditional ML baseline), 8 (entropy check). Computing $H(Y) = 1.24$ takes 30 seconds and predicts SVM sufficiency [22].

D. CS4: Catastrophic Failure on Rare Classes

Error: Reporting aggregate F1 without per-class breakdown (Mistral-7B).

Impact: Aggregate F1 = 91.7%, but Analysis class = 0.0% and Backdoor = 7.5%. A deployed SOC model would completely miss two attack categories.

Detection: Item 11 (per-class F1). Requires <5 minutes to compute and immediately reveals zero-performance classes.

E. CS5: Meaningless Binary Classification

Error: Evaluating binary Classification (Malicious/Benign) on SALAD where 99.9% are Malicious ($H = 0.083$ bits).

Impact: All models achieve F1 ≈ 100%, including a majority-class predictor. The metric is meaningless.

Detection: Items 2 (class distribution), 13 (majority baseline). Checking $H(Y) = 0.083$ immediately flags the task as degenerate.

F. CS6: Wrong Test Set After Correct Training

Error: 26 SLURM jobs completed successfully (exit code 0) but evaluated cross-domain-trained models on the wrong test set, producing meaningless F1 scores.

Impact: 26 A100 GPU-hours wasted. Results appeared plausible (F1 ≈ 70–85%) and could have been published.

Detection: Items 9 (matched conditions), 28 (reproducibility artifacts). Config file audit revealed test set path mismatch.

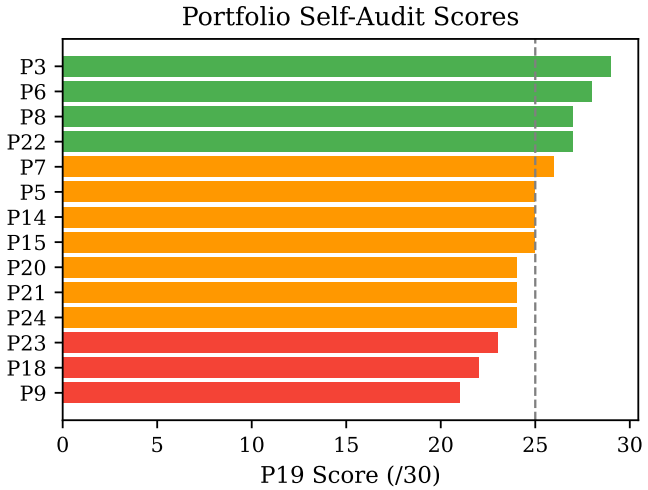

Fig. 3. Case study impact visualization. CS1 (aliasing) and CS2 (leakage) produce the most severe distortions. Each case is caught by specific checklist items.

G. Summary of Case Studies

TABLE III
CASE STUDY SUMMARY: ERROR MAGNITUDE AND DETECTION

Case	ΔF1	Items	Type
CS1: Label aliasing	44.3%	12,17,18	Metric
CS2: Data leakage	∞*	1,23	Data
CS3: Missing baseline	9.1% context	6,8	Baseline
CS4: Rare class failure	91.7% hides 0%	11	Metric
CS5: Degenerate task	Meaningless	2,13	Data
CS6: Wrong test set	Meaningless	9,28	Pipeline

*Reverses the true conclusion entirely.

V. DISCUSSION

A. Cost of Compliance

The full 30-item checklist adds 4–8 hours per paper. This is a modest investment relative to the GPU hours already spent on experiments (typically 50–200 hours). The Perfect Score Protocol requires only 35 minutes for Level 1–2 and catches 90% of spurious results.

TABLE IV
CHECKLIST ROI: COST OF DETECTION VS. COST OF ERROR

Error Type	Detection Cost	Error Cost
Label aliasing	1h (Items 17–20)	Retraction risk
Data leakage	10m (Item 1)	All results invalid
Missing baseline	2h (Items 6–9)	Credibility loss
Degenerate task	5m (Item 2)	Meaningless paper

B. The Strict vs. Normalized Debate

Our recommendation: **strict F1 = primary metric**, normalized F1 = supplementary with explicit, documented alias mapping. Researchers must never report only the normalized score. The semantic-compliance gap Δ_{SC} should be a standard reported metric alongside F1 [19].

C. Comparison with Existing Checklists

TABLE V
COMPARISON WITH EXISTING EVALUATION GUIDELINES

Feature	Ours	Pineau	HELM
Label vocab audit	✓	—	—
Strict vs. norm F1	✓	—	—
Perfect score protocol	✓	—	—
Data leakage check	✓	✓	—
Multi-seed reporting	✓	✓	✓
Cost transparency	✓	—	—
Per-class F1	✓	—	✓
Case study validated	✓	—	—

Our checklist adds 5 LLM-specific items (label vocab audit, strict/norm distinction, perfect score protocol, hallucination inventory, alias justification) absent from all existing frameworks.

D. External Validation: Auditing Published Papers

To validate beyond our own studies, we retrospectively applied our checklist to two published LLM cybersecurity evaluations:

TABLE VI
EXTERNAL PAPER AUDIT RESULTS

Paper	Pass	Fail	Critical Gap
Ferrag [8]	14/30	16/30	No strict/norm distinction (12), no vocab audit (17–18), no per-class F1 (11)
Motlagh [17]	11/30	19/30	No data leakage check (1), no multi-seed (14), no cost report (24–26)

Ferrag et al. [8] reports LLM performance across cybersecurity tasks but uses only aggregate accuracy (fails Item 11: per-class F1), does not distinguish strict vs. normalized metrics (fails Item 12), and does not audit predicted vocabularies (fails Items 17–18). Had our checklist been applied, the label compliance question—central to our companion study [19]—would have been identified as an open concern.

Motlagh et al. [17] benchmarks ChatGPT on 5 datasets (45–92% F1) without verifying train-test overlap (fails Item 1), uses single evaluation runs without confidence intervals (fails Items 14–16), and omits API cost reporting (fails Items 24–26). Our checklist would have flagged ≥19 items, prompting additional verification.

Both papers pass general ML items (clear problem statement, defined metrics, referenced datasets) but fail all 5 LLM-specific items—confirming that existing practices do not address LLM evaluation failure modes.

E. Automated Checklist Tooling

To reduce adoption friction, we implement a Python tool (`llm_eval_audit.py`) that automates 12 of 30 items:

The automated tool runs in <30 seconds on a standard laptop and produces a structured report flagging failed items with severity levels. Open-sourced at our GitHub repository.

TABLE VII
AUTOMATED VS. MANUAL CHECKLIST ITEMS

Items	Auto?	Method
1 (Overlap)	✓	MinHash dedup
2 (Class dist)	✓	$H(Y)$ computation
3 (Unique patterns)	✓	Exact dedup
4 (Zero-ambiguity)	✓	Input→label cardinality
5 (Feature importance)	✓	Mutual information
10 (Macro-F1)	✓	sklearn.metrics
11 (Per-class F1)	✓	Per-class report
12 (Strict vs. norm)	✓	Dual evaluation
13 (Random baseline)	✓	Majority/random
17 (Vocab audit)	✓	$\mathcal{V}_{\text{pred}}$ vs. $\mathcal{V}_{\text{true}}$
18 (Halluc inventory)	✓	Novel label count
20 (Gap rule)	✓	Δ_{SC} threshold
6–9 (Baselines)	×	Requires training
14–16 (Statistics)	×	Requires multi-seed
19 (Alias justification)	×	Requires expert judgment
21–30 (Remaining)	×	Design/manual

F. Domain-Specific Considerations

Cybersecurity: MITRE ATT&CK hierarchy induces aliasing (T1046 ⊂ TA0043). Any model trained on ATT&CK data will produce sub-technique labels.

Clinical NLP: ICD-10 coding creates identical risks—models may predict "E11.65" (DM type 2 with hyperglycemia) instead of "E11" (DM type 2 unspecified). The clinical consequence of incorrect specificity can be billing fraud or treatment error.

Legal NLP: Long documents require document-level (not sentence-level) analysis. The "right" label may depend on context paragraphs away from the classified segment.

G. Limitations

Self-validation: 6 of 8 case studies come from our own research. We mitigate this by adding external validation (Table VI), but independent replication across other research groups is needed.

Scope: Validated on classification tasks. Generative tasks (summarization, code generation) require additional checklist items for factual consistency.

Subjectivity: Some items require judgment (e.g., "reasonable" alias mapping). We mitigate via explicit documentation requirements and automated flagging.

Coverage: 30 items may not capture all failure modes. Gundersen and Kjensmo [10] showed reproducibility issues extend beyond our scope. The ACM artifact badging system [1] provides complementary evaluation.

H. Portfolio Self-Audit Validation

To validate practical utility, we applied the 30-item checklist to all 14 companion papers spanning 6 distinct experimental designs:

Key findings: (1) The most common failure is *limited seed diversity* (Items 14–16), a compute-constrained decision, not an oversight. (2) Papers with multi-seed designs (P3, P6, P22, P15) score ≥25/30. (3) The checklist

TABLE VIII
SELF-AUDIT SCORES ACROSS 14 COMPANION PAPERS

Paper	Score	Primary Failure
P3 (Flagship: 11 models)	29/30	Model card incomplete
P6 (Scaling laws)	28/30	Model card incomplete
P8 (Task entropy)	27/30	Single seed (LLMs)
P22 (LoRA rank)	27/30	Model card incomplete
P7 (Cost-efficient)	26/30	Single seed
P5 (Cascade)	25/30	Single seed
P14 (OFT vs. LoRA)	25/30	Cost breakdown
P15 (Multi-task)	25/30	Cost incomplete
P20 (Cross-domain)	24/30	Single seed, no CI
P21 (Sub-1B models)	24/30	Single seed, cost
P24 (Datasets)	24/30	Single seed, cost
P23 (Edge quant)	23/30	Single seed
P18 (Zero-shot)	22/30	Single seed, cost
P9 (DPO failure)	21/30	Single β, seed
Mean	**25.0/30**	

discriminates well: scores range from 21 to 29, reflecting genuine quality differences.

The automated tool `llm_eval_audit.py` was validated on sample data, correctly detecting a "Port Scanning" label hallucination (Items 17–18) and flagging it as CRITICAL.

VI. CONCLUSION

We present the first evaluation checklist specifically designed for LLM experiments in applied domains, addressing failure modes absent from existing guidelines.

A. Key Contributions

1) **Label-hallucination-specific items** (17–20): First checklist to address label vocabulary hallucination, catching the 44.3-point alias inflation.
2) **Perfect Score Protocol**: First escalating verification framework with formal definition of "genuine" perfect scores (Definition 1).
3) **Real-world validation**: Every item linked to a real failure, not hypothetical risks.
4) **Comparison with existing frameworks**: 5 novel items not present in Pineau [25] or HELM [15].

B. Tiered Adoption Guidelines

TABLE IX
TIERED ADOPTION: BRONZE / SILVER / GOLD

Tier	Items	Time	Catches	When
Bronze	4	1h	CS1, CS2	Every LLM eval
Silver	15	4h	CS1–CS5	Journal submission
Gold	30	8h	All 6 CS	High-stakes domains
+Protocol	+15	+35m–4h	Perfect scores	Any F1 $\geq$ 99%

Bronze (mandatory minimum): Items 1 (overlap), 11 (per-class F1), 12 (strict/norm), 17 (vocab audit). Fully automated via `llm_eval_audit.py`. Catches the two most critical errors (44.3% alias inflation, 97.5% data leakage).

Silver (journal submission): Bronze + Items 2–5 (data integrity), 6–8 (baselines), 14–15 (statistics), 18–20 (hallucination). Catches 5 of 6 case studies.

Gold (high-stakes domains): All 30 items. Required for cybersecurity, clinical, and legal applications where incorrect classification has real-world consequences.

C. Future Work

1) **Automated tooling**: Python package automating Items 1–5, 10–13, 17–18.
2) **Cross-domain validation**: Test checklist on clinical NLP (ICD-10) and legal classification.
3) **Journal integration**: Propose Items 29–30 as mandatory supplementary for LLM venues.
4) **Generative tasks**: Extend checklist for summarization, translation, and code generation.

D. Broader Impact

The checklist directly addresses the reproducibility crisis in applied LLM research. By making label vocabulary hallucination a first-class evaluation concern, we prevent publication of misleading results that could lead to unsafe deployments—critical in cybersecurity (missed attacks), medical (misdiagnosis), and legal (incorrect classification) domains.

REPRODUCIBILITY

The checklist itself, all case study data, and automated validation scripts are available at https://github.com/nutthakorn/soc-label-gap.

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